Artificial Intelligence: An Overview and History

From Turing's Vision to the Age of Large Language Models

What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and solve problems. Unlike traditional software that follows rigid, hand-coded rules, AI systems are designed to adapt based on data and experience. The field spans a wide range of capabilities, from recognising speech and images to playing complex games, translating languages, and generating human-like text.

AI is broadly categorised into two types. Narrow AI, also called weak AI, is designed to perform a specific task such as filtering spam emails or recommending movies. General AI, sometimes called strong AI or AGI (Artificial General Intelligence), refers to a hypothetical system with the reasoning and learning capabilities of a human across any domain. As of today, all deployed AI systems are narrow AI, though research into AGI is actively ongoing.

Machine learning is the most prominent subfield of AI. Rather than being explicitly programmed with rules, machine learning systems learn patterns from large datasets. Deep learning, a further subset, uses artificial neural networks with many layers to process complex data like images, audio, and text. It is deep learning that has driven most of the dramatic advances in AI over the past decade.

The History of Artificial Intelligence

1940s–1950s: The Founding Ideas

The conceptual roots of AI trace back to mathematician Alan Turing, who in 1950 published his landmark paper 'Computing Machinery and Intelligence,' introducing the famous question: 'Can machines think?' Turing proposed what became known as the

Turing Test — a benchmark for machine intelligence based on a machine's ability to exhibit intelligent behaviour indistinguishable from that of a human.

The field was formally established as a discipline in 1956 at the Dartmouth Conference, organised by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon. McCarthy coined the term 'Artificial Intelligence' at this conference, and the attendees believed that human-level machine intelligence could be achieved within a generation. That optimism would prove premature, but the conference launched decades of dedicated research.

1960s–1970s: Early Programs and the First AI Winter

The decades following Dartmouth saw the creation of early AI programs that could solve algebra problems, prove logical theorems, and play checkers. ELIZA, developed at MIT in 1966, was one of the first programs to simulate natural language conversation, serving as an early precursor to modern chatbots. These programs were impressive for their time but were brittle — they could only operate within extremely narrow, pre-defined boundaries.

By the mid-1970s, progress had stalled. Funding dried up as governments and institutions grew disillusioned with the gap between promises and results. This period became known as the first AI Winter — a phase of reduced interest, funding cuts, and widespread scepticism about the viability of AI research. The fundamental problem was that early AI lacked the data and computational power needed to scale.

1980s–1990s: Expert Systems and a Second Winter

The 1980s brought renewed enthusiasm through expert systems — AI programs that encoded the knowledge of human specialists in specific domains like medicine or engineering using a large set of if-then rules. Companies invested heavily, and expert systems were deployed in industries ranging from finance to manufacturing. Japan launched its ambitious Fifth Generation Computer Project to build intelligent machines, prompting the US and Europe to pour funding back into AI.

However, expert systems proved expensive to maintain, difficult to scale, and unable to handle situations outside their narrow training. The market collapsed in the early 1990s, leading to a second AI Winter. Meanwhile, in academia, quieter but more durable progress was being made. Researchers were developing backpropagation for training neural networks, and the foundations of modern machine learning were being laid.

2000s–2010s: The Deep Learning Revolution

The 2000s and early 2010s saw AI reborn through three converging forces: massive datasets generated by the internet, affordable GPU hardware capable of parallel computation, and algorithmic innovations in deep neural networks. In 2012, a deep learning model called AlexNet won the ImageNet image classification competition by a margin so decisive that it shocked the computer vision community and triggered a global shift toward deep learning.

This era produced a string of milestones. Google's AlphaGo defeated the world champion at the game of Go in 2016, a feat previously thought to be decades away. AI-powered speech recognition, machine translation, and image generation became mainstream. Tech giants including Google, Meta, Microsoft, and Amazon made AI the centrepiece of their research and product strategies, and the field attracted unprecedented levels of investment and talent.

The Era of Large Language Models

The most transformative recent development in AI has been the rise of large language models (LLMs). In 2017, researchers at Google introduced the Transformer architecture in the paper 'Attention Is All You Need.' The Transformer's ability to process entire sequences of text simultaneously, rather than word by word, allowed models to be trained at scales previously impossible. This architecture became the foundation of virtually all modern LLMs.

OpenAI's GPT series demonstrated that scaling up Transformers — training them on more data with more parameters — produced qualitative leaps in capability. GPT-3, released in 2020 with 175 billion parameters, could write essays, answer questions, summarise documents, and generate code with remarkable fluency. The release of ChatGPT in November 2022 brought these capabilities directly to the public, reaching 100 million users in just two months — the fastest adoption of any consumer technology in history.

Today, the landscape includes models from multiple organisations — GPT-4 from OpenAI, Claude from Anthropic, Gemini from Google, and open-weight models like Meta's LLaMA family. These models power a new generation of AI applications: coding assistants, document analysis tools, customer service agents, and retrieval-augmented generation (RAG) systems that can answer questions grounded in specific knowledge bases.

The challenges ahead are significant. Hallucination — where models generate confident but incorrect information — remains an open problem. Questions around bias, copyright, energy consumption, and the potential displacement of jobs are subjects of active debate among researchers, policymakers, and the public. Yet the pace of progress shows no signs of slowing, and artificial intelligence now sits at the centre of one of the most consequential technological transformations in human history.
